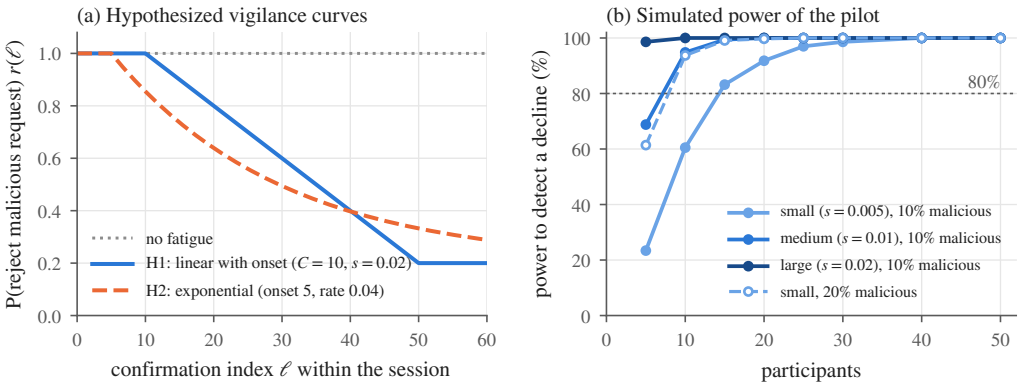
(a) Hypothesized vigilance curves
P(reject malicious request) $r(\ell)$
1.0
0.8
0.6
0.4
0.2
0.0
no fatigue
H1: linear with onset ($C = 10$, $s = 0.02$)
H2: exponential (onset 5, rate 0.04)
0
10
20
30
40
50
60
confirmation index $\ell$ within the session
(b) Simulated power of the pilot
power to detect a decline (%)
100
80
60
40
20
0
80%
small ($s = 0.005$), 10% malicious
medium ($s = 0.01$), 10% malicious
large ($s = 0.02$), 10% malicious
small, 20% malicious
0
10
20
30
40
50
participants